

FloorFlow: Discovering Operational Structure from Wireless Signals

Today, indoor positioning systems are built around a strong assumption: the building is already known.

Hospitals typically start with floor plans, room labels, department boundaries, and predefined zones. Technologies such as Wi-Fi, Bluetooth Low Energy (BLE), and Ultra-Wideband (UWB) are then used to answer a relatively narrow question: **Where is a person or piece of equipment within that known structure?**

Our project asks a different question:

What if the structure itself is unknown?

Imagine responding to an emergency in an unfamiliar hospital, a temporary field hospital, a large public shelter, or a building whose operational use has changed over time. In these situations, floor plans may be outdated, departments may have moved, and the way people actually use the building may differ significantly from its design.

Rather than treating localization as the final goal, we treat it as raw data.

Our system consumes wireless observations—whether from Wi-Fi, BLE, UWB, or future sensing technologies—and automatically discovers the building's operational structure. Without any floor plans, room labels, department information, or predefined zones, we identify:

- Natural zones formed by movement patterns and signal similarity
- High-traffic corridors and transition regions
- Bottlenecks and congestion points
- Areas where people tend to gather or wait
- Functional relationships between different parts of the building

The output is not simply a collection of coordinates. Instead, it is a learned graph of how the building actually operates.

This represents a shift from **localization** to **understanding**.

Historically, research has focused on improving positioning accuracy: better signal processing, better fingerprinting, denser Bluetooth deployments, and increasingly precise technologies such as UWB. As these technologies improve, obtaining location data becomes easier. However, the harder and more valuable problem becomes understanding what that location data means.

In fact, the value of our system increases as sensing technology improves.

With traditional BLE and Wi-Fi, signals are sparse and noisy. With future generations of UWB and wireless sensing, we can expect richer, denser, and more accurate movement data. Rather than making our project obsolete, these advances make it more powerful. Better sensing provides better observations, allowing our algorithms to build more accurate operational maps, discover more nuanced structures, and detect critical patterns earlier.

Our vision is a future in which buildings can automatically explain themselves.

Given only wireless observations, the system learns where the important spaces are, how they connect, and how people actually move through them. This capability could support emergency response, hospital operations, disaster management, crowd safety, and digital twin creation—without requiring prior knowledge of the environment.

In short, while existing systems ask:

"Where is this person?"

we ask:

"What is this building, and how does it function?"